Verification-First Legal RAG: Defense-in-Depth Auditing for Multilingual Swiss Law

Jonas Hertner[1]

[1]OpenCaseLaw — Independent researcher and maintainer, Zurich, Switzerland, team@jonashertner.com https://opencaselaw.ch

April 28, 2026

Abstract

Peer-reviewed measurement (Dahl et al., 2024; Magesh et al., 2024) finds that 58–82 % of legal queries to general-purpose LLMs and 17–33 % to commercial legal-RAG tools produce a fabricated authority. We present a five-rail closing audit pipeline for retrieval-augmented legal LLMs that addresses each error class with the cheapest mechanism that can detect it. The rails are: case-citation existence, statute-reference resolution, verbatim-quote source-matching, decision-date sanity, and proposition grounding by an independent LLM judge. Four rails are deterministic (∼3 ms per draft); the fifth is judge-mediated (∼2 s, ≈$0.005 per call). We release **Swiss Legal RAG Bench v0.2**, to our knowledge the first multilingual benchmark for legal-RAG *evaluation* on Swiss federal law (DE / FR / IT, 30 author-curated questions; every BGE evidence anchor resolved against the live corpus before publication; v1.0 will add named expert co-authors and a multi-annotator protocol). We stress-test the audit pipeline against the same generator in a prior-only condition (no retrieval) — chosen to maximise the density of recoverable errors, not as a substitute for end-to-end RAG evaluation — and measure: (i) **3.0 %** citation-level fabrication, well below Magesh's 17–33 % on commercial legal-RAG tools (one plausible explanation, untested here, is Swiss-corpus presence in the model's pre-training); and (ii) of the 6 wrong drafts, only 1 contains a fabricated citation — the remaining 5 are *real citation + misrepresented holding*. The case-existence rail catches the one fabricated draft at 0 % FPR; adding the other three deterministic rails preserves TPR (16.7 %) but raises FPR to 8.3 %; adding the second-pass LLM-judge grounding rail extends TPR to 50 % at 25 % FPR. The remaining 3 missed drafts are not judge-failures but audit-pipeline gaps: claim-extraction skipped the citation on two and the judge call returned an error on one. Closing these gaps is the open problem. Code MIT; benchmark CC0; underlying decision corpus consists of official published court decisions of their issuing courts, with packaging and added metadata licensed CC0. v0.2 and the audit pipeline are seeds for v1.0 (150 questions, named expert co-authors).

1 Introduction

Legal practice is unusually unforgiving of confident-sounding incorrect answers. A lawyer who cites a fabricated case, miscites a statute, or attaches a false holding to a real authority risks professional discipline and a wrong outcome for the client; in Switzerland, this is a violation of the *Sorgfaltspflicht* (duty of care) under the federal *Anwaltsgesetz* (BGFA, SR 935.61, art. 12). As large language models become a routine part of legal research, this asymmetry collides with a measured failure mode: 58–82 % of legal queries to general-purpose LLMs (Dahl et al., 2024) and 17–33 % of queries to commercial legal-RAG products (Magesh et al., 2024) produce a fabricated authority, miscited statute, or unsupported quotation.

Two recent developments shape the present work. First, Butler & Butler (2026) introduce a clean methodology for evaluating legal-RAG systems: decompose each error into three orthogonal binary signals (correctness c_{eli}, groundedness g_{eli}, retrieval accuracy r_{eli}) and read off the failure class from their joint distribution. Their methodology lets us ask *which* rail of a verification system addresses *which* error class. Second, the Swiss legal NLP literature has matured rapidly: SCALE (Rasiah et al., 2023) introduced a multi-task Swiss legal benchmark suite, the Niklaus group released anonymization tooling (Niklaus, Mamié et al., 2023) and re-identification risk measurement (Nyffenegger, Stürmer & Niklaus, 2024) for Swiss court rulings, SwiLTra-Bench (Niklaus et al., 2025) provided 180 k aligned Swiss legal translation pairs, and the SLDS dataset (Rolshoven, Rasiah et al., 2025) offered 20 k cross-lingual SFSC summarisation pairs. None of these target retrieval-augmented *generation* as the unit of evaluation; the present paper occupies that specific gap.

Two findings, in tension, motivate the rail decomposition. Our central empirical result is that on a closed Swiss-law corpus (969 k decisions, daily refresh, in the model's pre-training) Claude Sonnet 4.6 fabricates only **3.0 %** of emitted citations — an order of magnitude below the Magesh et al. (2024) 17–33 % measurement on commercial legal-RAG products. At face value this would suggest verification is unnecessary. But the c_{eli} correctness rate is only 80 % (24/30): of the 6 wrong drafts, only 1 contains a fabricated citation; the remaining 5 stem from a different mechanism — the model cites a *real* BGE *correctly* but *misrepresents* the holding, omitting the load-bearing precedential element or attributing a different doctrine to the case. Per-citation existence checks cannot detect this; an LLM-judge grounding rail extends recall but at meaningful false-positive cost. This residual class is what motivates the defense-in-depth design that follows.

The conceptual move. A production verification system for legal RAG should not be a single black-box LLM judge over the final draft, nor a single per-citation retrieval check, but a **decomposition into independent rails, one per named error class**, each implemented with the cheapest mechanism that can detect that class. The decomposition lets each rail be tested, ablated, and improved in isolation, and lets every draft pay only for the rails it needs. We instantiate the design as a five-rail closing audit pipeline (§3) deployed in production at `mcp.opencaselaw.ch`.

Contributions.

1. A defense-in-depth closing audit pipeline for legal RAG, with five rails addressing the four Magesh / Dahl error classes plus a fifth (date confabulation). Latency ≤ 3 ms for the four deterministic rails; $\sim$2 s, $\approx$\$0.005 per call for the LLM-judge rail.

2. **Swiss Legal RAG Bench v0.2**: to our knowledge the first multilingual benchmark for legal-RAG *evaluation* on Swiss federal law (DE / FR / IT, 30 author-curated questions), modelled on Butler & Butler (2026) with cross-lingual retrieval added as a fifth evaluation dimension. Released as a seed; v1.0 (150 questions, named expert co-authors, multi-annotator protocol) is forthcoming.

3. Per-rail ablation identifying two structural gaps in verification-first legal RAG: (a) the residual error class is *real citation, misrepresented holding* (5 of 6 wrong drafts in our experiment, vs. only 1 fabricated citation); (b) the audit's claim-extraction heuristic skips the citation on a non-trivial fraction of drafts, blocking the grounding judge before it can evaluate. In our experiment the judge converts 2 of 2 wrong drafts to issues whenever it gets to evaluate one, but is blocked on 3 of the 5 residual-class drafts (claim-extraction skips $\times 2$, judge call error $\times 1$).

2 Related Work

Legal-LLM hallucination measurement. Dahl et al. (2024, *Large Legal Fictions*) profiled hallucination on six legal-task types in general-purpose LLMs, reporting fabricated-authority rates of 58–82 %. The follow-up Magesh et al. (2024, *Hallucination-Free?*) measured commercial legal-RAG tools (Lexis+ AI, Westlaw AI-Assisted Research, Ask Practical Law) at 17–33 %. Our paper accepts these as the problem statement; the five-rail pipeline of §3 is the treatment.

Legal-RAG benchmarks. Butler & Butler (2026, *Legal RAG Bench*) introduced the four-leaf error decomposition methodology we extend. Their benchmark covers Australian criminal law (4,876 passages from the Victorian Criminal Charge Book, 100 hand-crafted questions) and uses a closed corpus. We adapt the methodology to Swiss multilingual federal law, adding cross-language retrieval as a fifth leaf because Switzerland is unusual among major jurisdictions in that the authoritative version of a leading case may be in a language other than the question's. Closed-domain multiple-choice benchmarks (LegalBench-RAG, Pipitone & Houir Alami 2024; HousingQA / BarExamQA, Zheng et al. 2025; LegalBench, Guha et al. 2023) do not test hallucination because the model never needs to invent an answer.

Swiss legal NLP. Recent Swiss legal NLP has matured rapidly around the Niklaus / Stürmer / Stanford / UZH / Bern axis. Three lines are directly adjacent to ours. (i) *Privacy and anonymisation*: Niklaus, Mamié et al. (2023) released BERT-based models for entity-level anonymisation of Swiss FSCS rulings, improving the operational *Anom2* system used by the Federal Supreme Court; Nyffenegger, Stürmer & Niklaus (2024) systematically measured LLM-based re-identification risk on anonymised Swiss court rulings, finding the risk minimal for top models on *court decisions* (higher on Wikipedia entities). We discuss the implications for our corpus exposure in §6. (ii) *Multilingual benchmarks*: SwiLTra-Bench (Niklaus et al., 2025) is a 180 k-pair Swiss legal translation benchmark across DE / FR / IT / RM / EN, introducing the SwiLTra-Judge specialised LLM evaluator with human-expert validation. SLDS (Rolshoven, Rasiah et al., 2025) provides 20 k SFSC rulings with cross-lingual headnote summarisations. We do not duplicate these tasks; our cross-lingual test (q-028) targets retrieval, not translation or summarisation. (iii) *Multi-task suites*: SCALE (Rasiah et al., 2023) covers citation extraction, court-view generation and summarisation; Swiss-Judgment-Prediction (Niklaus, Chalkidis & Stürmer, 2021) addresses outcome prediction; MultiLegalPile (Niklaus et al., 2023) provides a 689 GB multilingual legal pre-training corpus. None of these target retrieval-augmented *generation evaluation*; the present paper occupies that specific gap.

Verification methods. Self-RAG (Asai et al., 2023) and CRAG (Yan et al., 2024) propose general-domain self-verification or retrieval-correction; both treat verification as *pre-generation* (judge passage quality before generating). Our pipeline is *post-generation* verification: parse the generated draft, audit each emitted reference against the closed corpus, return a verdict per rail. The decomposition by named error class is, to our knowledge, novel for legal RAG; it is analogous in spirit to the SwiLTra-Judge calibration approach (Niklaus et al., 2025) but addresses retrieval-grounded text generation rather than translation evaluation.

Positioning. No prior work proposes a *deployed* mechanism that systematically reduces hallucination by decomposing the verification problem into independent error-class detectors. Existing RAG systems either treat verification as a single LLM-judge pass over the final output (which inherits the same model's biases) or as per-citation retrieval validation (which catches existence errors but not propositional mis-grounding). The five-rail closing audit we propose in §3 is the first to address each error class with the cheapest mechanism that can detect it, with each rail independently testable and ablatable.

3 The Five-Rail Audit Pipeline

The audit operates on a draft answer D and a retrieval-state pool S (decisions and statutes the generator has access to). It returns a list of issues $\mathcal{I}(D, S)$, each labelled with one of five categories (Figure 1). An OK verdict (no issues) means the audit found nothing; ¬OK means at least one rail fired. The five rails operate *in parallel* on the same draft.

LLM draft D $\Longrightarrow$

Rail 1: case-citation existence (~ 2 ms, deterministic)
Rail 2: statute-reference resolution (~ 1 ms, deterministic)
Rail 3: verbatim-quote source-matching (~ 2 ms, deterministic)
Rail 4: decision-date sanity (< 1 ms, deterministic)
Rail 5: proposition grounding (opt-in) (~ 2 s, judge)

$\Longrightarrow$ **verdict** $\mathcal{I}(D, S)$

Figure 1: Five-rail closing audit pipeline. Four deterministic rails address citation-, statute-, quotation-, and date-level fabrications; an opt-in LLM-judge rail addresses proposition-grounding errors. Each rail is independently testable and ablatable; the configuration adopted by a deployment is a per-rail subset.

Rail 1 — case-citation existence. A regex parses the draft for Swiss case-citation forms (BGE / ATF / DTF, BGer / TF dockets, BVGer / BStGer / BPatGer dockets, MKGE numerical references). Each parsed citation is resolved against the corpus by exact-match lookup with a strict canonical-prefix guard; the LIKE-fallback path is suppressed for inputs that look like canonical decision IDs because such inputs either hit the indexed exact-match path or are fabricated. Latency: 2–5 ms per draft regardless of citation count. Catches: invented BGE numbers, mis-typed docket numbers, citations to retracted decisions. Optional pinpoint check: if the citation includes E. $n.m$ or consid. $n.m$, the pinpoint must match either the structured-extraction sidecar or the body-text `regeste` of the cited decision.

Rail 2 — statute-reference resolution. A second regex parses for Art. N LAW forms (with German / French / Italian abbreviation synonyms) and resolves each against the Fedlex SPARQL mirror cached locally as `statutes.db`. The resolution is two-step — abbreviation → SR number → article text — and is cached at process scope. Latency: 1–3 ms per draft. Catches: fabricated law abbreviations, wrong article numbers, references to repealed articles. Connector words ("und", "oder", "et", "ou") are case-sensitively filtered to prevent the regex eating them as fake law codes after Abs. N structures.

Rail 3 — verbatim-quote source-matching. Quoted substrings of ≥ 30 characters in the draft are extracted (German „...“, French « ... », Italian « ... », English curly " . . . ", ASCII "...") and required to appear verbatim — after whitespace + quote-mark normalisation — in the source pool S. The pool comprises the `regeste` text plus first 8 k characters of each cited decision, plus the article text of each cited statute. Latency: ∼2 ms. Catches: hallucinated quotations, paraphrases inside quotation marks.

Rail 4 — decision-date sanity. For each verified case citation, parse adjacent vom / *du* / *del* DD.MM.YYYY constructs within 60 characters of the citation span and compare to the stored `decision_date`. Catches: model emits a real BGE but pairs it with a wrong date, suggesting confusion between two cases with similar dockets. Latency: < 1 ms.

Rail 5 — proposition grounding (opt-in). For each verified citation, the immediately preceding claim sentence is extracted (up to 250 characters before the citation, terminated by sentence boundary; "see"-cite markers "vgl." / "siehe" / "cf." / "cfr." suppress the rail). All (claim, source) pairs go in a *single batched* prompt to a separate Sonnet judge call (same model family as the generator; see §6) that returns a JSON array of verdicts {`yes, partial, no, contradicts, unrelated`}. Verdicts in {`no, contradicts, unrelated`} become issues. One Sonnet call per draft regardless of citation count. Latency: 1.5–2.5 s. Cost: ≈ $0.005 per call. Catches: real citation + misrepresented holding (the Butler / Butler "reasoning error" class).

Output canonicalisation. When all rails pass, the audit emits a `linked_text` field with every parsed citation replaced by its corpus-canonical form wrapped in a Markdown link to the source URL on `mcp.opencaselaw.ch/entscheid/<id>`. Drafts arrive with messy citation spacing or case and leave publication-ready.

4 Swiss Legal RAG Bench v0.2

We release Swiss Legal RAG Bench v0.2 alongside this paper. Version 0.2 contains 30 questions *author-curated by the maintainer of* `opencaselaw.ch` *(the corpus mirror underlying the audit pipeline)*, drawn from the major substantive areas of Swiss federal law (Schuldrecht, Sachenrecht, Familienrecht, Erbrecht, Strafrecht, Strafprozess, Verfassungsrecht, Verwaltungsrecht, Mietrecht, Arbeitsrecht, Sozialversicherungsrecht, Verkehrsrecht, Aktienrecht, SchKG, Datenschutz, Verjährung, Wirtschaftsfreiheit; 17 distinct areas). The language distribution is DE = 18, FR = 8, IT = 4 (60 / 27 / 13 %). v0.2 is explicitly framed as a single-curator seed; v1.0 will adopt a multi-annotator protocol with named expert co-authors from Swiss law faculties (see §6).

Question construction. Each question is paired with a concise reference answer (1–4 sentences in the question's language) and annotated supporting evidence (statute SR-number + article number + language; leading-case `decision_id` (s) where applicable). Of the 30 questions, 16 (53 %) are anchored on a verified leading-case BGE; 14 (47 %) carry the `doctrine_from_bge` claim type, in which the expected answer derives a doctrinal proposition from a seed BGE rather than restating its holding directly. The two annotations are not disjoint: every `doctrine_from_bge` question is also anchored on a leading-case BGE. The split exists to let the ablation surface where each rail contributes — case-existence on the seed citations, the grounding judge on the doctrinal proposition.

Corpus verification of evidence. Every `decision_id` cited as evidence was resolved against the live corpus (`mcp.opencaselaw.ch/api/decisions/<id>`) before the benchmark was published; every BGE returned a 200 response with non-zero citation count and an on-topic `regeste`. This is itself a methodological discipline: most legal-RAG benchmarks build evidence annotations from secondary sources and inherit any errors there. We commit to corpus-first construction, with the audit trail published in the supplementary material.

Cross-lingual extension. Switzerland is unusual among major civil-law jurisdictions in that the authoritative version of a leading case may be in a language other than the question's. Question q-028 (IT) anchors on DTF 126 III 113 (which has regesti in DE / FR / IT) to test the cross-language retrieval claim. v1.0 expands this dimension to a full cross-language retrieval suite.

Versioning. v0.2 is a seed sufficient to characterise the audit pipeline in this paper. **v1.0** is targeted at 150 questions with named expert co-authors from Swiss law faculties and a multi-annotator protocol (inter-annotator agreement reported, third-party adjudication for disagreements). We explicitly invite contributors for v1.0.

5 Experiments

Setup. We use Claude Sonnet 4.6 (released 2026-02-17) as both the generator and the judge for c_{eli}, g_{eli}, and the in-pipeline grounding rail. Using one model family on both sides is a known limitation (§6); v1.0 will report inter-judge agreement against an independent model. All 30 questions in v0.2 are run in a *prior-only* condition (no retrieval): the generator sees only the question text and must rely on training knowledge. We chose the prior-only condition as an audit-pipeline *stress test*, not as an end-to-end RAG evaluation: it maximises the density of recoverable errors and is the closest replication of Dahl et al. 2024's general-LLM measurement. A retrieval-augmented condition runs in production at `mcp.opencaselaw.ch` (BM25 / RRF / Haiku rerank, see §6); end-to-end RAG evaluation against v1.0 is forthcoming work.

For each question we collect: the prior-only draft, the per-category issue counts from `attest_response` with all five rails on, the c_{eli} judge verdict against the v0.2 reference answer, and the g_{eli} judge verdict against the empty source pool. g_{eli} is uniformly 0 in the prior-only condition by construction (no retrieved sources $\Rightarrow$ no claim can be grounded). We use the c_{eli} verdict as ground truth for the per-rail ablation.

Per question we issue exactly four Sonnet calls (one generation, one in-pipeline grounding judge, one c_{eli} judge, one g_{eli} judge) plus one deterministic-rails pass; the 6-row ablation in Table 1 is computed in post-processing by intersecting each draft's per-rail fire flags with each configuration's enabled rail set. Total: 120 Sonnet calls across 30 drafts; 106 s wall-clock with 4-way parallelism; $<$\$2 in API charges.

Citation-level fabrication rate. Across 30 prior-only drafts, Claude Sonnet 4.6 emits 33 case citations of which 32 resolve in the corpus. The citation-level fabrication rate is **3.0 %**, well below Magesh et al.'s 17–33 % on commercial legal-RAG tools and Dahl et al.'s 58–82 % on general LLMs. Two non-exclusive hypotheses for the gap, neither tested in the present work: (i) Swiss BGE references appear at scale on the open web — including via our own published Hugging Face dataset — and the model may have internalised the citation conventions in pre-training; (ii) the prior-only condition asks short, well-bounded questions where the model can refuse, hedge, or ground in widely-known leading cases, whereas Magesh's tools were exercised on more open-ended litigation queries. We do not claim the same rate would obtain for *any* model on *any* legal corpus, nor that the contrast against Magesh generalises beyond Swiss law and Sonnet 4.6.

Per-rail ablation. Of the 30 prior-only drafts, 24 are judged correct ($c_{eli} = 1$) and 6 wrong ($c_{eli} = 0$) by the second-pass Sonnet judge (same model family as the generator; see §6). Table 1 reports the catch rate of each cumulative configuration on the wrong-draft subset (true positives) and the correct-draft subset (false positives).

Three findings emerge. First, the deterministic rails behave very differently from one another: only the case-existence rail (configuration C_1) holds the perfect-precision floor (0 % FPR), and it does so by catching a single fabrication, q-017's invented DTF 117 II 198 (which Sonnet uses to anchor an otherwise-correct paraphrase of art. 264 CO). Adding the statute, quote, and date rails (C_2–C_4) preserves TPR but degrades precision: FPR climbs to 8.3 % by C_4 as the rails flag valid statute references and quotations as anomalies on correctly-answered drafts. Second, the grounding rail (C_5) is responsible for the entire jump in TPR over C_4: it catches q-018 and q-030, the only two wrong drafts on which the rail's claim-extraction heuristic identified a preceding claim sentence and the judge call returned (i.e. TPR conditional on the judge actually evaluating the draft is 2/2). The 25 % FPR is the cost of that recall: six correct drafts trigger the rail because the second-pass judge disagrees with the draft's interpretation of a real cited authority. Third, the three wrong drafts that all rails miss (q-021, q-025, q-026) are missed for a structural reason different from the explanation we originally hypothesised.

Table 1: Per-rail-configuration catch rates, v0.2 prior-only condition. $n = 30$ drafts; $c_{eli} = 0$ for 6 (wrong) and $c_{eli} = 1$ for 24 (correct) per the second-pass Sonnet judge. TPR = $P(\text{any enabled rail fires} \mid c_{eli} = 0)$, computed on $n = 6$; FPR = $P(\text{any enabled rail fires} \mid c_{eli} = 1)$, computed on $n = 24$. Net = TPR − FPR (Youden's J). 95 % Wilson confidence intervals on TPR are wide because of the small wrong-draft denominator ($3/6 \to [18.8, 81.2]$); v1.0 will tighten these with ≥ 30 wrong drafts in a 150-question benchmark.

Config	Rails enabled	TPR	TPR 95% CI	FPR	Net
C_0	$\emptyset$ (no rails)	$0/6 = 0.0\,\%$	$[0.0, 39.0]$	0.0 %	+0.0 pp
C_1	+ case	$1/6 = 16.7\,\%$	$[3.0, 56.4]$	0.0 %	+16.7 pp
C_2	+ statute	$1/6 = 16.7\,\%$	$[3.0, 56.4]$	4.2 %	+12.5 pp
C_3	+ quote	$1/6 = 16.7\,\%$	$[3.0, 56.4]$	8.3 %	+8.3 pp
C_4	+ date (4 deterministic)	$1/6 = 16.7\,\%$	$[3.0, 56.4]$	8.3 %	+8.3 pp
C_5	+ grounding (full 5-rail)	$3/6 = 50.0\,\%$	$[18.8, 81.2]$	25.0 %	+25.0 pp

The residual error class. The three missed drafts (q-021, q-025, q-026) share a substantive pattern — Sonnet cites a real BGE (BGE 142 III 433, BGE 144 III 481, BGE 130 V 343) correctly but *mischaracterises* the holding, omitting the load-bearing precedential element — which we call the *real-citation / misrepresented-holding* class. However, the proximate reason these drafts escape the audit is *not* that the grounding judge evaluates them and concurs with Sonnet's mischaracterisation: it is that the grounding rail never gets to evaluate them. Per `grounding_meta` in the artifact: q-021 and q-026 trip the `skipped_no_claim` branch (the heuristic that extracts the claim sentence preceding a citation found no candidate within its 250-character window), and q-025 returns a `judge_unavailable` error (a transient API failure during the experimental run). The residual error class is therefore really two failures stacked on top of each other: an audit-pipeline gap (claim-extraction heuristic, judge-call reliability) prevents the verifier from *evaluating* the drafts that contain the most subtle error type, and the substantive error type itself is the hardest one even given evaluation. Closing the audit gap is mechanical (more permissive claim extraction, retry-on-error) and is the obvious next deployment step; testing whether the grounding judge actually catches misrepresented-holding errors when it gets to evaluate them requires v1.0's larger denominator and an independent judge model.

Cost / latency. Across the 30 drafts, the four deterministic rails together account for ∼3 ms per draft (median 2.7 ms; 95 % under 6 ms), measured by wrapping the rail dispatch in `time.perf_counter_ns()` on the production deployment. The grounding rail takes 1.5–2.5 s wall-clock and ≈$0.005 per draft via the Anthropic API. Total experimental cost (generation + audit + judges): <$2.

6 Discussion

When the deterministic rails miss. The case-citation rail catches *fabricated* references, not *misused* ones. For systems where the underlying model has high citation-level fidelity (as Sonnet 4.6 has on Swiss law: 3.0 % fabrication rate), the deterministic rails contribute most of their value as a precision floor: they make sure the remaining 97 % of valid-looking citations are in fact valid. Of the four deterministic rails, only C_1 (case-existence) holds 0 % FPR in our experiment; adding statute, quote, and date raises FPR to 8.3 %. A production deployment can run all four cheaply, but the precision-floor argument for running them on every draft applies cleanly only to C_1 as currently implemented; the other three need to be tightened or made user-suppressible before they fire as freely.

When the judge rail miscalls. The 25 % FPR of the grounding rail in our experiment overstates the production rate, because v0.2's prior-only condition gives the judge little context. With retrieval, the judge sees the actual cited text and can ground its verdict properly. However, the fundamental tradeoff remains: a stronger judge catches more reasoning errors but also raises more disagreement-as-error verdicts. Calibrating the judge — when to defer to the draft, when to override — is open work.

Generalisation. Each rail is corpus-agnostic. The case-existence rail needs an enumerable decision corpus; the statute rail needs a statute mirror; the quote rail needs the cited decisions' verbatim text. None of these are Swiss-specific. An English-law adaptation (e.g. England & Wales) would require: (i) regex patterns for English neutral citations and ICLR references; (ii) a statute mirror for primary legislation (`legislation.gov.uk`-derived); (iii) decision full-text for the quote rail. All three are public-domain artefacts. We expect the catch-rate / FPR profile to be model-dependent: a weaker generator with higher citation-level fabrication rate will see higher rail TPR.

Privacy and re-identification. The corpus we publish (969 k Swiss court decisions, CC0) consists exclusively of decisions *as published by the issuing court* or via the official Swiss judicial publication portal `entscheidsuche.ch`. We perform no de-anonymisation, no re-identification, and no enrichment with non-public data. Nyffenegger, Stürmer & Niklaus (2024) measured LLM-based re-identification risk on anonymised Swiss court rulings and found it minimal for top frontier models on *court decisions* specifically (substantially higher on Wikipedia entities); we adopt their finding and discuss it explicitly in the dataset card. Our contribution to privacy is one-direction: we expose the already-public anonymised corpus through a more uniform interface, which does not create re-identification capacity beyond what was already feasible by direct scraping of the source portals. We do *not* mirror the in-court (pre-anonymisation) corpus; that work — including the operational *Anom2* system — belongs to the Niklaus, Mamié et al. (2023) line and the Federal Supreme Court itself.

Limitations.

- **Benchmark size.** v0.2 is a 30-question seed. With only 6 wrong drafts, the per-configuration TPR estimates are imprecise: 95 % Wilson confidence intervals on a 50 % point estimate ($n = 6$) are $[18.8\%, 81.2\%]$. We report exact ratios in Table 1 so the reader can judge directly. v1.0 (150 questions, named expert co-authors from Swiss law faculties; see §4) is the planned next deliverable.
- **Single-curator benchmark.** v0.2's questions and reference answers were authored by the maintainer of `opencaselaw.ch`, a Swiss-trained author with practical legal-research experience but without the multi-annotator protocol that a published gold-standard benchmark requires. Selection bias is the obvious risk: a curator who built the underlying corpus may unintentionally choose questions whose evidence the corpus answers well. We mitigate by anchoring every evidence `decision_id` on the live corpus before publication and by publishing the evidence-resolution audit trail; we cannot eliminate the bias, and v1.0 addresses it directly with named expert co-authors and an independent annotation protocol.
- **Same model family, generator and judge.** Both roles are filled by Claude Sonnet 4.6 in our experiment. This concentrates failure modes (a wrong draft and a wrong-but-self-consistent judge will agree). v1.0 will adopt a SwiLTra-Judge-style independent calibrated judge (Niklaus et al., 2025) with human-expert validation, and report inter-judge agreement.

- **Prior-only condition.** Our reported ablation runs the generator without retrieval, chosen as a stress-test of the audit pipeline (it maximises the density of recoverable errors), not as an end-to-end RAG evaluation. Butler & Butler (2026) report that retrieval is the primary determinant of legal-RAG performance; we do not contradict them, and an end-to-end retrieval-augmented evaluation against v1.0 is forthcoming. The audit pipeline is run on every production query at `mcp.opencaselaw.ch` (over a BM25 / RRF / Haiku-rerank retriever), but production-side audit catch-rates are not the subject of the present paper.
- **Single-run experiment.** The numbers in Table 1 reflect one run; we do not bootstrap. Run-to-run variance at `temperature=0` is small but non-zero.

7 Conclusion

We propose a defense-in-depth closing audit pipeline for legal RAG, decomposing verification into five rails with bounded latency and cost. We release Swiss Legal RAG Bench v0.2 as, to our knowledge, the first multilingual benchmark for legal-RAG evaluation on Swiss federal law. In our prior-only stress test, the case-existence rail catches the one fabricated citation among 6 wrong drafts at 0 % FPR; the second-pass grounding judge catches a further 2 of the 5 *real-citation, misrepresented-holding* drafts (it scores 2/2 when the audit's claim-extraction heuristic gets it a claim to evaluate); 3 wrong drafts escape the audit because the claim-extraction heuristic skipped ($\times 2$) or the judge call returned an error ($\times 1$) before any verdict was reached. Closing those audit-pipeline gaps and stress-testing the judge on a larger-denominator v1.0 against an independent model are the next steps.

The audit pipeline is live at `mcp.opencaselaw.ch`, deployed on the entire published Swiss federal and cantonal court corpus (969,000+ decisions, daily refresh). Code under MIT; benchmark packaging under CC0; underlying decision texts are official published court decisions of their issuing courts (with the collation, deduplication, and added metadata licensed CC0). We invite collaborators for v1.0.

Data and Code Availability

Reproducibility pin (2026-04-28). The numbers in §5 were produced from these exact artefacts:

- Audit pipeline source (MIT), pinned to the experiment commit:
  `https://github.com/jonashertner/caselaw-repo-1/blob/4553379/mcp_server.py`
- Benchmark v0.2 (CC0): `https://huggingface.co/datasets/voilaj/swiss-legal-rag-bench`
  SHA-256 of `questions.jsonl`: `b299db37...f21ec449`
- Per-trial ablation data, pinned to the experiment commit:
  `https://github.com/jonashertner/caselaw-repo-1/blob/4553379/docs/paper/v2/experiments/ablation_results.json`
  SHA-256: `79dcc442...039c3b08`
- Underlying Swiss-caselaw corpus (packaging and added metadata under CC0; underlying texts are official published court decisions of the issuing courts): `https://huggingface.co/datasets/voilaj/swiss-caselaw`
  Live MCP corpus size at experiment time: 969,738 decisions
- Live MCP endpoint (for re-execution against current corpus): `https://mcp.opencaselaw.ch`

Generator and judge: `claude-sonnet-4-6` (Anthropic), `temperature=0`. Each trial in §5 can be re-executed by pinning to the commits + SHAs above and pointing the harness at the live MCP endpoint.

No competing financial interests declared. This work was funded by the author and the volunteer OpenCaseLaw contributor community. The author maintains the project as a public-interest infrastructure; no commercial product is sold.

References

[1] A. Asai, Z. Wu, Y. Wang, A. Sil, H. Hajishirzi. Self-RAG: Learning to retrieve, generate, and critique through self-reflection. arXiv:2310.11511, 2023.

[2] A.-R. Butler, U. Butler. Legal RAG Bench: an end-to-end benchmark for legal RAG. Isaacus, March 2026. arXiv:2603.01710.

[3] M. Dahl, V. Magesh, M. Suzgun, D. E. Ho. Large Legal Fictions: Profiling Legal Hallucinations in Large Language Models. *Journal of Legal Analysis*, 16(1), 64–93, 2024. arXiv:2401.01301.

[4] M. Geering, J. Merane. Swiss Federal Supreme Court Dataset, 2024. Hugging Face datasets repository, `rcds/swiss_court_view_generation`.

[5] N. Guha, J. Nyarko, D. E. Ho, et al. LegalBench: A collaboratively built benchmark for measuring legal reasoning in large language models, 2023.

[6] V. Magesh, F. Surani, M. Dahl, M. Suzgun, C. D. Manning, D. E. Ho. Hallucination-Free? Assessing the Reliability of Leading AI Legal Research Tools. Stanford RegLab working paper, 2024 (arXiv:2405.20362); peer-reviewed version forthcoming in *Journal of Empirical Legal Studies*, 2025.

[7] J. Niklaus, I. Chalkidis, M. Stürmer. Swiss-Judgment-Prediction: A Multilingual Legal Judgment Prediction Benchmark, 2021.

[8] J. Niklaus, V. Matoshi, M. Stürmer, I. Chalkidis, D. E. Ho. MultiLegalPile: A 689 GB Multilingual Legal Corpus, 2023.

[9] J. Niklaus, R. Mamié, M. Stürmer, D. Brunner, M. Gygli. Automatic Anonymization of Swiss Federal Supreme Court Rulings. In *Proceedings of the Natural Legal Language Processing Workshop 2023*, ACL.

[10] A. Nyffenegger, M. Stürmer, J. Niklaus. Anonymity at Risk? Assessing Re-Identification Capabilities of Large Language Models in Court Decisions. In *Findings of NAACL 2024*, pages 2433–2462.

[11] J. Niklaus, J. Merane, L. Nenadic, S. Ahmadi, Y. Gao, et al. SwiLTra-Bench: The Swiss Legal Translation Benchmark. In *Proceedings of ACL 2025*, pages 14894–14916.

[12] L. Rolshoven, V. Rasiah, S. Brügger Bose, S. Hostettler, L. Burkhalter, M. Stürmer, J. Niklaus. Unlocking Legal Knowledge: A Multilingual Dataset for Judicial Summarization in Switzerland. In *Findings of EMNLP 2025*, pages 15382–15411.

[13] N. Pipitone, G. Houir Alami. LegalBench-RAG: A benchmark for retrieval-augmented generation in the legal domain. arXiv:2408.10343, 2024.

[14] V. Rasiah, R. Stern, V. Matoshi, M. Stürmer, I. Chalkidis, D. E. Ho, J. Niklaus. SCALE: Scaling up the Complexity for Advanced Language Model Evaluation, 2023.

[15] S. Yan, J. Gu, Y. Zhu, Z. Ling. Corrective Retrieval Augmented Generation. arXiv:2401.15884, 2024.

[16] L. Zheng, N. Guha, J. Arifov, S. Zhang, M. Skreta, C. D. Manning, P. Henderson, D. E. Ho. A reasoning-focused legal retrieval benchmark. In *Proceedings of the Symposium on Computer Science and Law*, ACM, 2025.